CHAPTER 3

Limits of functions

S21, S23, S24, S26

Before you begin · Limits of functions

Without a calculator, in six minutes. Answer briefly. Use sign, factorisation and roots. Then choose the targeted follow-up indicated under the questions.

1. Factorise $x^2 - 4$.
2. Give the sign of $\frac{x-1}{x+2}$ on $]-2;1[$.
3. Simplify $\sqrt{x^2}$ for $x < 0$.
4. What is the limit of the sequence $\frac{1}{n+1}$?
5. Determine the dominant term of 3 $x^3 - x$.
6. Give an excluded value for $\frac{1}{x-2}$.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Expand (x-2)(x+2) to check. **2.** Make two sign rows, then their quotient. **3.** Test x=-3 then generalise; R28. **4.** Revisit the unit on limits of sequences and the band around 0. **5.** Compare the degrees before any simplification. **6.** Find where the denominator is zero before calculating. The R questions serve as materials for review with the teacher, even if their collective routine is planned for later.

Objectives

Calculate limits at a point and at infinity; resolve an indeterminate form; interpret an asymptote.

Course notes

Meaning of a limit

$f(x) \to \ell$ when $x \to +\infty$ means that $f(x)$ can be made as close as desired to ℓ for all x sufficiently large. For a limit at a, consider x of the domain, close to a and distinct from a. Left and right limits are written $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$. When the domain has points on both sides of a, a two-sided limit exists if and only if the two one-sided limits exist and are equal. At an endpoint of the domain, we study the limit on the only accessible side.

Graphical interpretation

If $f(x) \to \ell \in \mathbb{R}$ when $x \to +\infty$ or $-\infty$, the line $y = \ell$ is a horizontal asymptote in this direction. If $f(x) \to \pm\infty$ when $x \to a$ on one side, the line $x = a$ is a vertical asymptote.

A curve can cross a horizontal asymptote. The limit does not necessarily give the position of the curve relative to the asymptote: study the sign of $f(x) - \ell$.

References and operations

At $+\infty$, $x^m \to +\infty$ for $m \geq 1$, $\frac{1}{x^m} \to 0$, $\sqrt{x} \to +\infty$. At $-\infty$, the sign of x^m depends on the parity of m. The rules for sum, product, quotient and comparison are those for sequences, with the same indeterminate forms.

Example. For $f(x) = \frac{2x+1}{x-3}$, write $f(x) = 2 + \frac{7}{x-3}$. The limits at $\pm\infty$ are equal to 2. At 3^-, $f \to -\infty$; at 3^+, $f \to +\infty$. The asymptotes are $x = 3$ and $y = 2$.

Conjugate. For $x > 0$, $\sqrt{x^2+4x} - x = \frac{4}{\sqrt{1+\frac{4}{x}}+1} \to 2$. For $x < 0$, be careful: $\sqrt{x^2} = |x| = -x$.

Exponential and comparative growth

$$\lim_{x\to+\infty} e^x = +\infty, \quad \lim_{x\to-\infty} e^x = 0, \quad \lim_{x\to+\infty} \frac{e^x}{x^m} = +\infty \quad (m \in \mathbb{N}).$$

We deduce $x^m e^{-x} \to 0$ at $+\infty$.

Proof of the limits of the exponential. The study of $e^x - x - 1$ gives $e^x \geq 1 + x$. By comparison, $e^x \to +\infty$ at $+\infty$. Setting $t = -x \to +\infty$, $e^x = \frac{1}{e^t} \to 0$.

Proof of a comparative growth result. For every integer $k \geq 0$ and $x \geq 0$, $e^x \geq \frac{x^k}{k!}$. For $k = 0$, $e^x \geq 1$. If the inequality holds at order k, the derivative of $e^x - \frac{x^{k+1}}{(k+1)!}$ is $e^x - \frac{x^k}{k!} \geq 0$; its value at 0 is positive. The inductive step follows. By choosing $k = m + 1$, $\frac{e^x}{x^m} \geq \frac{x}{(m+1)!} \to +\infty$.

Choose a transformation

Quotient of polynomials: factor out the highest powers. Difference of square roots: multiply by the conjugate. Quotient near a zero: factorise and study the signs. Exponential versus power: comparative growth, after a change of variable if necessary.

Comparison and composition: conditions of use

If $g(x) \leq f(x) \leq h(x)$ near the point or infinity studied, and if g, h have the same real limit ℓ, then $f(x) \to \ell$ (squeeze theorem). If $g \leq f$ and $g \to +\infty$, then $f \to +\infty$. For a limit $-\infty$, we use an upper bound by a function tending to $-\infty$. If $u(x) \to a$ and v is continuous at a, then $v(u(x)) \to v(a)$, when the composite is defined. For example $\sqrt{1+\frac{1}{x}} \to 1$ at $+\infty$. For an infinite limit of the outer function, check from which side $u(x)$ approaches the bound: $\frac{1}{u(x)}$ does not have the same behaviour according to the sign of $u(x)$.

Read a limit, choose a transformation

3.1 The approach side is part of the question

First use: S23.

Worked example · A vertical asymptote

For $f(x) = 1 + \frac{1}{x-2}$, the domain is $\mathbb{R} \setminus \{2\}$. When $x \to 2^-$, the denominator is negative and tends to 0, so $f(x) \to -\infty$. When $x \to 2^+$, $f(x) \to +\infty$. The line $x = 2$ is a vertical asymptote; no value $f(2)$ exists. At infinity, $f(x) \to 1$.

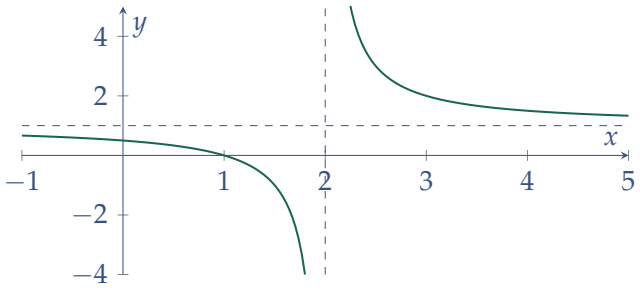


Figure 3.1. Find the two one-sided limits on the drawing: the branches are not joined.

A1 · Immediate application Determine the limits of $g(x) = 3 - \frac{2}{x+1}$ at -1^-, -1^+, $-\infty$ & $+\infty$. Give the asymptotes.

3.2 An indeterminate form calls for a calculation

First use: S24.

Worked example · The sign hidden in a square root

For $x < 0$, $\sqrt{x^2} = |x| = -x$. Thus, at $-\infty$,

$$\frac{\sqrt{x^2+1}}{x} = \frac{|x|\sqrt{1+1/x^2}}{x} = -\sqrt{1+1/x^2} \longrightarrow -1.$$

At $+\infty$, the same expression tends to 1. Omitting the absolute value would produce a positive limit for an expression that is actually negative.

A2 · Fading guidance Calculate at $+\infty$ the limits of $\frac{2x^2+1}{x^2-x}$, $\sqrt{x^2+4x} - x$ and $x\left(\sqrt{1+1/x} - 1\right)$. For the first, we suggest dividing by x^2 ; choose freely for the others.

A3 · Construct examples Construct three quotients whose limit has the form $0/0$ at 0, whose limits are respectively 0, 1 and $+\infty$. State the side of approach if necessary.

3.3 Comparative growth and change of variable

First use: S26; logarithm after S49.

Property · Exponential and powers · results from the lesson

For every integer $m \geq 1$, $e^x/x^m \to +\infty$ & $x^m e^{-x} \to 0$ at $+\infty$. To transfer these results, identify the new variable and its limit.

Worked example · Transferring a result

When $x \to -\infty$, set $t = -2x$ gives $t \to +\infty$ & $x^2e^{2x} = \frac{1}{4}t^2e^{-t} \to 0$. After S49, the substitution $t = 1/x$ in $x \ln x$ for $x \to 0^+$ gives $-(\ln t)/t \to 0$. The logarithm is not used before it has been studied.

A4 · Deferred application After S26, calculate $\lim_{x\to+\infty}(x^2+3x)e^{-x}$ & $\lim_{x\to-\infty} xe^x$. Say why formally replacing x by infinity is not a justification.

3.4 An asymptote may be crossed

First use: S24.

Definition · Horizontal Asymptote

If $f(x) - \ell \to 0$ at one of the infinities, $y = \ell$ is an asymptote at this infinity. This definition requires a vertical distance that tends to 0; it does not rule out any intersection at a finite distance.

Worked example · Two behaviours

On $\mathbb{R}$, $p(x) = 1 + 1/(1+x^2)$ remains strictly above $y = 1$. The function $q(x) = 1 + x/(1+x^2)$ crosses this same line at $x = 0$, while tending to 1 at both infinities. For q, the sign of $q(x) - 1$ is that of x.

A5 · Refute and explain A student says that the curve of q cannot cross its asymptote. Give the coordinates of the point of intersection and the signs on either side. Is this consistent with the two limits?

A6 · Choice of strategy Determine a, b so that $h(x) = \frac{ax+b}{x-1}$ has the horizontal asymptote $y = 2$ and satisfies $h(0) = 3$. Then study the one-sided limits at 1 and the position of the curve relative to $y = 2$.

One minute, spoken aloud. For each of the expressions in the "Methods" exercise, first state what prevents a conclusion from the operations on limits, then explain the role of the transformation.

Delayed review · S49 then S81. Redo the change of variable with $\ln x / x$; sort again the limits, mixing logarithms, exponentials and roots.

Investigation

A curve close to a line - 30 min.

Consider the function f defined on $\mathbb{R} \setminus \{1\}$ by $f(x) = \frac{3x-2}{x-1}$. Let $\mathcal{C}_f$ its curve in a coordinate system of the plane.

1. Show that, for every $x \neq 1$, $f(x) = 3 + \frac{1}{x-1}$.

2. Determine the limits of f at $-\infty$, at 1^-, at 1^+ and at $+\infty$.

3. Study the sign of $f(x) - 3$, then specify the position of $\mathcal{C}_f$ with respect to the line with equation $y = 3$.

4. Determine a real number $A > 1$ such that, for every $x > A$, $|f(x) - 3| < 0{,}01$.

5. Does curve $\mathcal{C}_f$ intersect its horizontal asymptote? Justify and explain why an overly wide graphing window can give a misleading impression.

Progressive exercises

E1 · Direct application. Determine $\lim\limits_{x \to +\infty} \dfrac{5x^3 - x}{2x^3 + 7}$ justifying.

E2 · Application and justification. Let f the function defined on $\mathbb{R} \setminus \{2\}$ by $f(x) = \frac{1}{(x-2)^2}$. Determine its limits at 2^-, at 2^+, at $-\infty$ and at $+\infty$.

E3 · Method to organise. Determine $\lim\limits_{x \to 1} \dfrac{x^2 - 1}{x - 1}$ specifying why the simplification used is valid near 1.

E4 · Choice of strategy. Determine $\lim\limits_{x \to +\infty} x^3 e^{-2x}$. State the comparative growth result used.

E5 · Synthesis or proof. Determine $\lim\limits_{x \to -\infty} \left(\sqrt{x^2 + 1} + x\right)$. Justify in particular the treatment of $\sqrt{x^2}$.

Work from the textbook or a past paper

Barbazo, p. 96, Exercises 26 and 29. In S23: exercise 29, to link limits and asymptotes. At home after S24: exercise 26, to construct three counterexamples to a faulty calculation rule on limits.

Homework

DM3 - A family of curves - 45 min.

For every real number a, consider the function f_a defined by $f_a(x) = \frac{ax^2+1}{x^2+1}$. Let $\mathcal{C}_a$ be its graph.

1. Determine the domain of f_a, then its limits at $-\infty$ and at $+\infty$.

2. Show that $f_a(x) - a = \frac{1-a}{x^2+1}$. Depending on the value of a, state the position of $\mathcal{C}_a$ with respect to the line with equation $y = a$.

3. Study the variations of f_a distinguishing the cases $a < 1$, $a = 1$ and $a > 1$. State any extrema and where they are attained.

4. Take $a = 3$. Solve, for $x \geq 0$, the inequality $|f_3(x) - 3| \leq 0{,}02$.

MCQ A · one-sided limits and transformations

After S24; MCQ B after S26.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. $\lim_{x\to 2^-} \frac{1}{x-2}$ is:

A. $+\infty$ B. $-\infty$

C. 0 D. 2

2. For $x < 0$, $\sqrt{x^2}$ is:

A. x B. x^2

C. $-x$ D. $\pm x$

3. $\lim_{x\to+\infty} \frac{3x^2-x}{x^2+4}$ is:

A. 0 B. $+\infty$

C. -1 D. 3

4. $\lim_{x\to 1} \frac{x^2-1}{x-1}$ is:

A. 2 B. 0

C. $+\infty$ D. does not exist

5. For $f(x) = 1 + \frac{\sin x}{x}$, $x > 0$, the line $y = 1$:

A. is not an asymptote B. is an asymptote with no intersection

C. is an asymptote crossed infinitely often D. is tangent at every point

MCQ B · comparative growth and choice of tool

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. $\lim_{x\to+\infty} x^2 e^{-x}$ is:

A. $+\infty$ B. 0

C. 1 D. $-\infty$

7. $\lim_{x\to-\infty} xe^x$ is:

A. 0 B. $+\infty$

C. $-\infty$ D. 1

8. The appropriate tool for $\sqrt{x^2+2x} - x$ at $+\infty$ is:

A. the binomial B. the IVT

C. differentiation D. the conjugate

9. If $f(x) \to 0$ and $g(x) \to +\infty$the product:

A. always tends to 0 B. always tends to $+\infty$

C. needs further study D. is impossible

10. If $f(x) \geq x^2$ for $x \geq 1$, then:

A. $f(x)/x^2 \to 1$ B. $f(x) \to +\infty$

C. f is increasing D. $f(x) - x^2 \to 0$

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. A graph never crosses its horizontal asymptote.

2. A finite limit at a requires that $f(a)$ is defined.

3. The symbol ∞/∞ gives a limit equal to 1.